

Milestone 2: Frequency Analysis of Musical Instruments

EE/CE 252/251 - Spring 2026

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Outline

- 1 Introduction
- 2 Individual Instrument Analysis (Part a)
 - Tabla
 - Hiba B6
 - Guitar C String
- 3 Corruption & Recovery (extension of Task A)
- 4 Comparative Analysis (Part b)
- 5 Synthesizer Implementation (Part c)
- 6 Detection System Design (Part d)
- 7 Conclusion

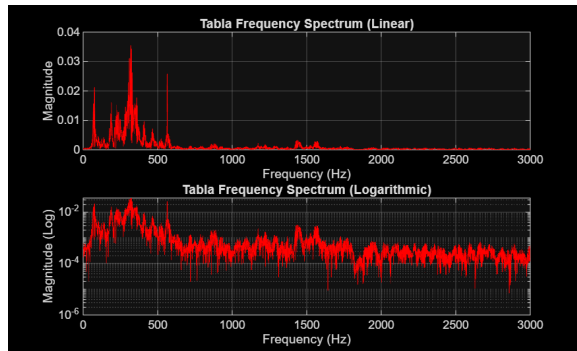
Project Objective

- Extend frequency-domain investigation from Task A to three self-recorded instruments:
 - **Tabla** (percussion)
 - **Female voice B6**
 - **Guitar C String**
- Specific goals (Task B parts a–d):
 - a) Individual harmonic analysis
 - b) Compare all five instruments (cello, trombone + three new)
 - c) Discuss synthesizer synthesis from keynote/overtones
 - d) Propose an automatic instrument detection system

- **Data acquisition:** recorded with smartphones, loaded via audioread
- **Tabla segment:** 1 s steady-state slice from middle of 5 s recording
- **FFT analysis:** single-sided amplitude spectrum (0–3000 Hz)
- **Peak detection:** findpeaks with 10% threshold
- **Visualization:** linear + logarithmic plots; comparative overlay

Tabla – Percussion (Maaz Haris)

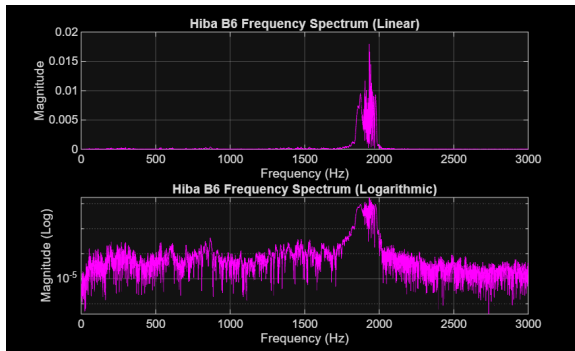
- **Duration:** 1 s (full 5 s)
- **Keynote:** 76.00 Hz
- **Overtone ratios:** non-integer (1.33, 1.80, 2.49 ...)
- **No. of overtones (log):** 13 distinct peaks
- **Noise:** 65.00 Hz (electrical interference)



Membrane inharmonicity prevents exact integer multiples.

Female Voice B6 (Basil Saeed Bari)

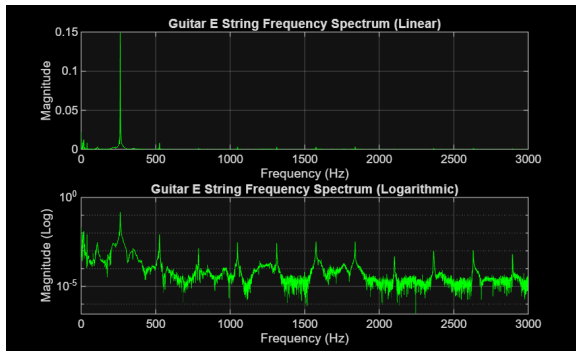
- **Duration:** 4.01 s
- **Keynote:** 1874.67 Hz (slightly flat)
- **2nd peak:** 1932.98 Hz (ratio 1.03)
- **Formant structure:** vocal tract shapes harmonic amplitudes
- **Noise:** negligible 54 Hz peak



Voice exhibits a rich formant envelope beyond integer multiples.

Guitar C String – Effective Keynote 262 Hz (Rauf Abdullah)

- **Duration:** 2.01 s
- **Keynote:** 262.22 Hz (C4)
- **Overtone ratios:** nearly exact 2, 3, 4, 5 ...
- **No. of overtones:** ~ 10 visible
- **Noise:** 64.99 Hz (mains interference)



Slight inharmonicity from string stiffness, similar to cello.

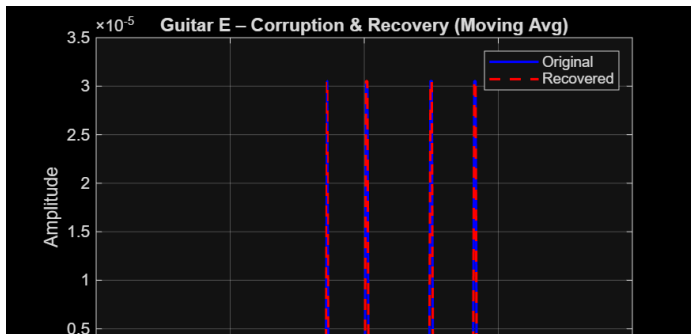
Moving-Average Corruption on Guitar

- Applied trombone-style 2-point moving average:

$$y[n] = \frac{1}{2}(x[n] + x[n-1])$$

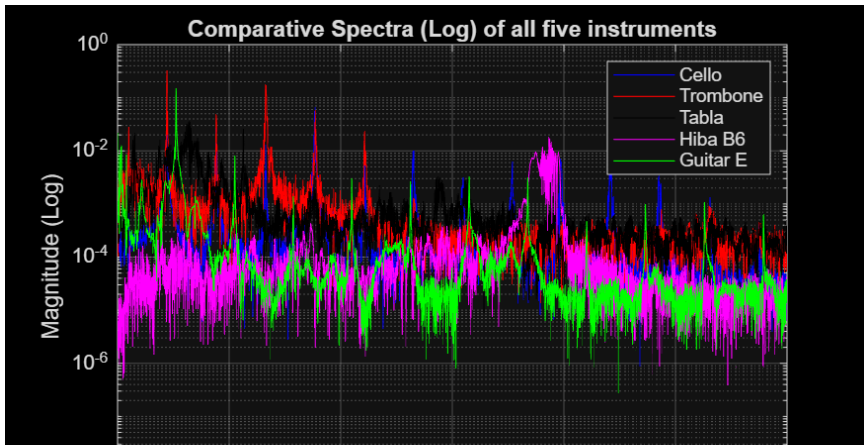
- Exact recovery possible because no samples discarded
- Inverse difference equation:

$$x[n] = 2y[n] - x[n-1], \quad x[1] = 2y[1]$$



Similarities Across Five Instruments

- All produce a fundamental + overtone series
- String instruments (cello, guitar) show inharmonicity; tabla membrane even stronger
- Logarithmic scale essential to reveal high-frequency content



Differences Across Five Instruments

- **Timbre:** relative overtone amplitudes differ drastically – tabla dense, trombone sparse, voice formant-shaped
- **Frequency spacing:** trombone near-perfect multiples; tabla/strings deviate
 $\rightarrow f_k = kf_0\sqrt{1+Bk^2}$
- **Noise signatures:** tabla & guitar: ~ 65 Hz hum; cello/trombone: tape-motor noise; voice: cleanest

Additive Synthesis from Spectral Data

- Extract f_0 , f_k , amplitudes A_k , phases ϕ_k from FFT
- Generate:

$$x_{\text{synth}}(t) = \sum_{k=1}^K A_k \cos(2\pi f_k t + \phi_k)$$

- Model inharmonicity: $f_k = kf_0 \sqrt{1 + Bk^2}$ (estimate B from data)
- Apply temporal envelope (ADSR): e.g., sharp attack + fast decay for tabla, slow vibrato for voice

Automatic Instrument Recognition

- ① Compute single-sided spectrum of a short segment
 - ② Find fundamental frequency (lowest significant peak)
 - ③ Extract normalised overtone amplitudes (first $N = 10$)
 - ④ Compare with template database (Euclidean / cosine distance)
 - ⑤ Classify to nearest instrument
- Robust because inharmonicity + amplitude ratios are unique fingerprints
 - Can be enhanced with temporal features (attack time) or neural network

Summary

- Three new instruments successfully analyzed in frequency domain
- Keynote frequencies, inharmonicity, noise artefacts documented
- Five-instrument comparison reveals distinct spectral fingerprints
- Additive synthesis can recreate realistic timbres
- Spectral pattern matching gives a practical detection framework
- Frequency-domain tools are essential to understand and exploit acoustic signatures

Thank you

Questions?